

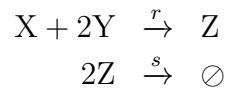
NETWORKS AND COMPLEXITY

Solution 12-2

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 12.2: Mass action laws [Lvl. 1]

Write the differential equations for x , y and z that correspond to the reaction diagram



Solution

The first reaction occurs proportional to r , x and y^2 . It decreases x by one unit, y by two units and increases z by one unit. So taking only this reaction into account would give us

$$\dot{x} = -rxy^2 \quad (1)$$

$$\dot{y} = -2rxy^2 \quad (2)$$

$$\dot{z} = rxy^2 \quad (3)$$

The second reaction occurs proportional to s and z^2 and reduces z by two units. Adding the term from this reaction we find the final result

$$\dot{x} = -rxy^2 \quad (4)$$

$$\dot{y} = -2rxy^2 \quad (5)$$

$$\dot{z} = rxy^2 - 2sz^2. \quad (6)$$